

Computer Circuit

Condensed Study Notes

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An Introduction to Computer Circuit

Imagine a computer's speed as being determined by how fast its internal pathways can send signals. To make a computer faster, we can either speed up how quickly its tiny switches turn on and off, or create pathways that can handle these faster signals.

In a more technical sense, a computer circuit is like a network of interconnected paths that electricity (specifically, electrons) can flow through. Think of it like a road system for tiny electrical cars.

These circuits operate on a simple, two-state system: either 'on' or 'off'. This is like a light switch – it's either letting electricity through or it's not.

To achieve these 'on' and 'off' states, computers use tiny electronic components called **transistors**. These transistors act like incredibly fast light switches, opening and closing in practically no time at all – billionths and even trillionths of a second! This speed is what allows computers to perform calculations so rapidly.

In theoretical computer science, a **circuit** is a way to model how calculations happen. It's like a flow chart where information (called **input values**) goes through a series of steps, with each step performing a specific task or calculation. These are similar to, and generalize, **Boolean circuits**, which are mathematical models for how digital logic works. These circuits are defined by the types of calculations (**gates**) they can perform and the kinds of answers (**values**) they can produce. For example, in a Boolean circuit, the answers are either 'true' or 'false', and the calculations include 'and', 'or', and 'not' operations.

Motherboard as the main circuit board in a PC

Imagine your computer as a bustling city. The motherboard is like the city's main hub, the central square where all the important buildings and roads connect. It's a flat board covered in pathways, like tiny roads, that allow different parts of the computer to talk to each other.

This main circuit board, called a **printed circuit board (PCB)**, is found in almost all computers, like desktops and laptops. It's the central point where everything plugs in – both the internal parts and the things you connect from the outside.

Think about the different parts inside your computer:

- **Chipsets:** These are like the traffic controllers of the city, managing the flow of information between different components.
- **Central Processing Unit (CPU):** This is the city's main processing center, where all the thinking and calculations happen. We'll learn more about this later!
- **Memory:** This is like the city's short-term memory storage, holding information the CPU needs quickly.

It also connects to things you see on the outside, like:

- **Wi-Fi and Ethernet:** These are like the city's communication lines for getting information from the internet.
- **Graphics Cards (with a Graphics Processing Unit or GPU):** This is like the city's art department, responsible for creating and displaying everything you see on your screen.

IC as the major Circuitry Technology in Modern PC

We've talked about computer circuits being like pathways inside a computer and the motherboard being like a city's main square connecting everything. Now, let's zoom in on a super important building block within those circuits: the **Integrated Circuit (IC)**.

Think of an IC as a tiny, self-contained city on its own, packed with miniature versions of all the components we talked about before – like transistors (those on-off switches) and resistors (which control the flow of electricity). These ICs are also called **microchips**.

Instead of having lots of separate little parts wired together, an IC puts hundreds of thousands of these components onto a single, small piece of material, often **silicon** (a type of material often used for computer parts). This makes them super compact and efficient.

Because they can hold so many components, ICs are capable of doing really complex jobs. They have different connection points, called **terminals**, which allow them to perform various functions, especially in things like **embedded systems** (which are computer systems built into other devices, like a car's computer).

There are tons of different types of ICs, designed for all sorts of purposes. For example, some are used for storing information (like **memory chips** in your computer), while others are found in cars to control different functions.

What is an Integrated Circuit

We've learned that computer circuits are like tiny pathways inside a computer, and that Integrated Circuits (ICs), or microchips, are like miniature cities packed with many of these pathways and components. Now, let's dive a bit deeper into what exactly an IC is!

Imagine a tiny, flat piece of special material, like a miniature building block. On this block, we can fit lots and lots of tiny electronic parts, like little rooms or specialized workshops, all connected together. This is essentially what an Integrated Circuit (IC) is.

More technically, an IC is a circuit where many electronic parts are built very close together and are permanently connected, working as one single unit to do many jobs.

These ICs are made on a single, flat piece of material called a semiconductor. On this semiconductor, we can build very tiny versions of components called MOSFETs (Metal Oxide Semiconductor Field Effect Transistors). Think of these MOSFETs as super-fast on-off switches, like the ones we discussed earlier, but made incredibly small.

Putting all these components together on one chip makes the computer:

- **Faster:** Signals don't have to travel as far.
- **More Reliable:** Fewer separate connections mean fewer things can go wrong.
- **More Efficient:** They use less power.
- **Less Expensive:** Mass production makes them cheaper to make.

Based on how they are built and what they do, ICs are generally divided into three main types:

- **Digital Integrated Circuits:** These work with information that is either 'on' or 'off', like a light switch. They are great for tasks like counting or making decisions.
- **Analog Integrated Circuits:** These work with signals that can change smoothly over a range, like the volume control on a radio. They are good for things like amplifying sound or processing signals from sensors.
- **Mixed Integrated Circuits:** As the name suggests, these combine both digital and analog parts on the same chip, allowing them to handle a wider variety of tasks.